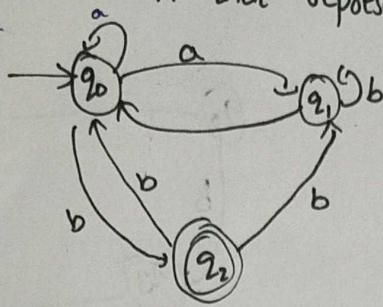


consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L



- States : $\{q_0, q_1, q_2\}$
- Initial state : q_0
- Final state : q_2
- Alphabet (Σ) : $\{a, b\}$

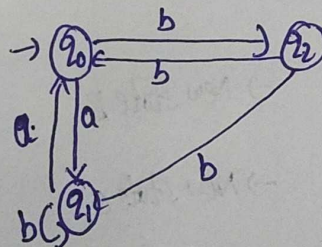
NFA Transition Table

	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
$*q_2$	$\emptyset$	$\{q_0, q_1\}$

NFA transition table

- $\delta(q_0, a) = \{q_0, q_1\}$
- $\delta(q_0, b) = \{q_2\}$
- $\delta(q_1, a) = \{q_0\}$
- $\delta(q_1, b) = \{q_1\}$
- $\delta(q_2, a) = \emptyset$
- $\delta(q_2, b) = \{q_0, q_1\}$

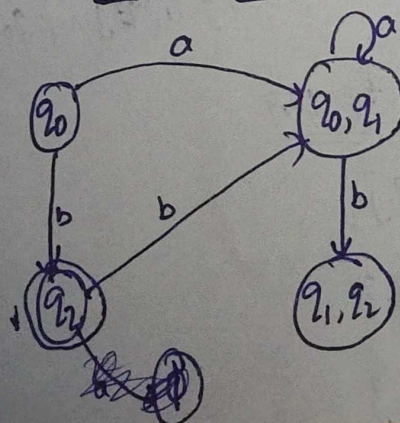
(dfa diagram)



DFA transition table

	a	b
q_0	$[q_0, q_1]$	$[q_2]^*$
$[q_0, q_1]$	$[q_0, q_1]$	$[q_1, q_2]^b$
$[q_2]^*$	$[\emptyset]$	$[q_0, q_1]$
$[q_0, q_1]^*$	$[q_0]$	$[q_0, q_1]$
$\emptyset$	$\emptyset$	$\emptyset$

DFA Diagram

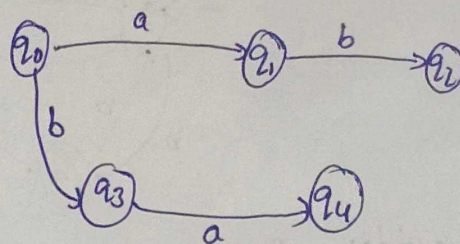


states : $\{q_0, q_1, q_2, q_3, q_4\}$

Initial state : q_0

Final state : q_2, q_4

NFA diagram



state	a	b
q_0	$\{q_0, q_1\}$	$\{q_0, q_3\}$
q_1	$\{\phi\}$	$\{q_2\}$
q_2	ϕ	ϕ
q_3	$\{q_4\}$	ϕ
q_4	ϕ	ϕ

Converting NFA to DFA

$$\delta(q_0, a) = \{q_0, q_1\} \rightarrow \text{New state B}$$

$$\delta(q_0, b) = \{q_0, q_3\} \rightarrow \text{New state C}$$

$$\delta(q_0, a) \cup \delta(q_1, a) = \{q_0, q_1\} \rightarrow B$$

$$\delta(q_0, b) \cup \delta(q_1, b) = \{q_0, q_2\} = \text{New state D}$$

$$\delta(q_0, a) \cup \delta(q_2, a) = \{q_0, q_1, q_2\} \rightarrow \text{New state E}$$

Transition table for DFA

$$A = \{q_0, q_1\}$$

$$B = \{q_0, q_2\}$$

$$C = \{q_0, q_3\}$$

$$D = \{q_0, q_2, q_4\}$$

$$E = \{q_0, q_1, q_2\}$$

DFA state	a	b
→ A	B	C
B	B	D
C	E	C
*D	E	C
*E	B	D

③ start state : q_0

Final state : ~~q~~ F

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

~~q₀ → start~~ $\delta \rightarrow$ transition function

state	0	1
→ q_0	$\{q_0, q_1\}$	$\{q_0, q_3\}$
q_1	$\{q_1\}$	$\emptyset$
* q_2	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\{q_4\}$
* q_4	$\emptyset$	$\emptyset$

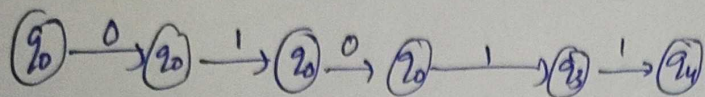
• Initial state : $\{q_0\}$

• Read 0 : $\delta(q_0, 0) = \{q_0, q_1\}$

• Read 1 : $\delta(q_0, 1) \cup \delta(q_1, 1) = \{q_0, q_3\}$

• Read 0 : $\delta(q_0, 0) \cup \delta(q_3, 0) = \{q_0, q_1\}$

- Read 1 : $\delta(q_0, 1) \cup \delta(q_1, 1) = \{q_0, q_3\}$
- Read 1 : $\delta(q_0, 1) \cup \delta(q_3, 1) = \{q_0, q_3, q_4\}$



The string 01011 is successfully accepted.

$Q : \{q_0, q_1, q_2, q_3\}$

$\Sigma : \{0, 1\}$

q_0 : start state

q_3 : Final state

state	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\emptyset$
q_2	$\emptyset$	q_3
$* q_3$	$\{q_3\}$	$\{q_3\}$

1. state $A = \{q_0\}$

$\delta_D(A, 0) = \delta_N(q_0, 0) = \{q_0, q_1\}$
 $\rightarrow$ New state B

2. process state $B = \{q_0, q_1\}$

$\{q_0, q_1, q_2\} \rightarrow$ New state C

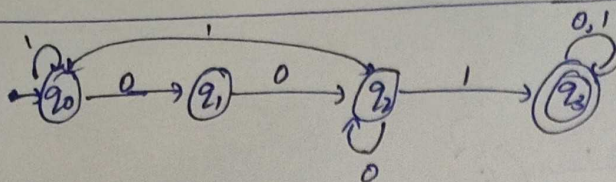
3. process state $C = \{q_0, q_1, q_2\}$

$\rightarrow$ New state D

DFA table :-

state	NFA Subset	0	1
$\rightarrow A$	$\{q_0\}$	B	A
B	$\{q_0, q_1\}$	C	A
C	$\{q_0, q_1, q_2\}$	C	D
$* D$	$\{q_0, q_3\}$	E	D

$* F \{q_0, q_1, q_3\}$



Finding ϵ -closure for each state

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

Step 1: $\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$

Input 0:

$$\delta_N(q_0, 0) \cup \delta_N(q_1, 0) \cup \{\delta_N(q_2, 0)\} = \emptyset$$

$$\{q_1, q_2\} \rightarrow \text{New state } E$$

Input 1:

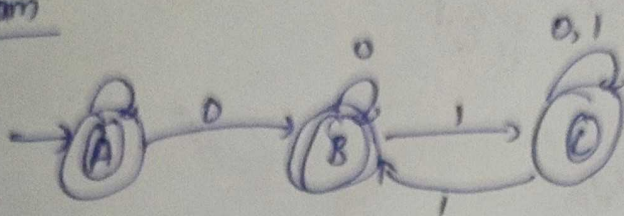
$$\delta_N(q_0, 1) \cup \delta_N(q_1, 1) \cup \delta_N(q_2, 1)$$

$$\epsilon\text{-closure}(\{q_0, q_2\}) = \{q_0, q_1, q_2\} = A$$

DFA table:

$\rightarrow$ DFA	NFA State Subsets	0	1
$\rightarrow$ *A	$\{q_0, q_1, q_2\}$	B	A
*B	$\{q_1, q_2\}$	B	C
*C	$\{q_2\}$	C	C

DFA Diagram



SIMATS ENGINEERING ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

91/180 J.